Democratization theory

A game-theoretic approach

Lecture 06

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Table of contents

1. Introduction

2. Aggregating Preferences

3. Single-Peaked Preferences

4. Median Voter Theorem

5. The Model

- **Approach:**

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  - Preference based individuals

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- Focus in this class: aggregating preferences and social choice

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  - What are reasonable properties of these preferences?
- How do we deal with these problems?

Individual Preferences

Let X be a choice set.

A *preference* $\succcurlyeq$ is a relation in X, i.e., $\succcurlyeq \subset X \times X$.

So, for $x, y \in X$, we can write $x \succcurlyeq y$ meaning that x is at least as good as y.

Rational preference: $\succcurlyeq$ is a rational choice if

- *(complete)* $\forall x, y \in X$, $x \succcurlyeq y$ or $y \succcurlyeq x$
- *(transitive)* $\forall x, y, z \in X$, if $x \succcurlyeq y$ and $y \succcurlyeq z$, then $x \succcurlyeq z$

A little on choice theory[1]

Definition: A function $u : X \longrightarrow \mathbb{R}$ represents a preference $\succcurlyeq$ when for all $x, y \in X$, $x \succcurlyeq y \Leftrightarrow u(x) \geq u(y)$

Theorem: If a preference $\succcurlyeq$ can be represented by a utility function, then $\succcurlyeq$ is rational.

What about the other way around?

[1] see more in https://impa.br/wp-content/uploads/2017/04/25CBM_08.pdf

A little on choice theory

Definition (monotonicity): Let $X = \mathbb{R}^n_+$. A preference over X is monotone if $\forall x, y \in X$ that $x \geq y$ and $x \neq y$ implies $x \succ y$, where $x \geq y$ iff $x_k \geq y_k$ for $k = \{1, 2, ..., n\}$.

Theorem: Let $X = \mathbb{R}^n_+$ and $\succcurlyeq$ be a preference rational and monotone over X. Then there is a utility function $u : X \longrightarrow \mathbb{R}$ that represents $\succcurlyeq$.

Remark: $x \succ y$ means that $x \succcurlyeq y$ and $\neg(y \succcurlyeq x)$

Concorcet Paradox [2]

Suppose we have three projects, A, B, and C, and three voters with preferences as follows

$A \succcurlyeq_1 B \succcurlyeq_1 C$

$B \succcurlyeq_2 C \succcurlyeq_2 A$

$C \succcurlyeq_3 A \succcurlyeq_3 B$

Note that all preferences are rational.

[2] https://en.wikipedia.org/wiki/Marquis_de_Condorcet

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- Which project will be chosen?

- **May's [3] Theorem:** A group decision function with an odd number of voters meets anonymous, neutral, and positively responsive if and only if it is the simple majority method.

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- Even more: these is the key result that creates a *conflict of interest*

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- With the notation of the Acemoglu and Robinson (2006) in mind, let Q be a set of all possible policy choices and $V^i : Q \longrightarrow \mathbb{R}$ be the utility function o individual i over Q[6]

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- In the model, $Q = [0, 1]$ representing an income tax and will be denoted by $q^i = \tau^i$

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Definition (Single-Peaked Preferences): Policy preferences of voter i are single-peaked if nad only if:

$$q'' < q' < q^i \text{ or } q'' > q' > q^i, \text{ then } V^i(q'') < V^i(q')$$

Strict concavity of $V^i(q)$ is sufficient for it to be single-peaked

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- open agenda: any individual can propose a new pairwise vote pitting any alternative against the winner from the previous vote

Median Voter Theorem

Theorem: Consider a set of policy choices $Q \subset \mathbb{R}$; let $q \in Q$ be a policy and let M be the median voter with ideal point q^M. If all individuals have single-peaked preferences over Q, then (1) q^M always defeats any other alternative $q' \in Q$ with $q' \neq q^M$ in a pairwise vote; (2) q^M is the winner in a direct democracy with an open agenda.

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- government redistribute equally money from tax: that means that lump-sum transfer is $T = (\tau - C(\tau))\bar{y}$

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- Hypothesis guarantee that V is concave on τ
- Problem of individual i

$$\tau^i = \arg\max_{\tau \in [0,1]} V(y^i|\tau)$$

Therefore, the first order condition to the individual i is given by

$$-y^i + (1 - C'(\tau))\bar{y} = 0 \text{ and } \tau > 0$$

or

$$-y^i + (1 - C'(\tau))\bar{y} \leq 0 \text{ and } \tau = 0$$

This last inequality means that $\tau = 0$ if $y^i \leq \bar{y}$.[7]

[7] Suppose $y^i = \bar{y}$. Then firs order condition implies that $C'(\tau^i) = 0$, which implies that $\tau^i = 0$

Implications: Using the implicit function theorem, we have

$$\frac{d(\tau(y^i))}{d(y^i)} = -\frac{1}{C''(\tau(y^i))\bar{y}} < 0$$

Tagging the median voter as M, the tax chosen by this voter, τ^M, will be given by

$$\frac{y^M}{\bar{y}} = 1 - C'(\tau^M)$$

Notice that the lower the income of the median voter in respect to the mean income, the higher the tax rate chosen by individual M

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- $y^p = \frac{(1-\theta)\bar{y}}{1-\delta}$ and $y^r = \frac{\theta\bar{y}}{\delta}$
- Restrictions: $\delta < 1/2$, $y^p < \bar{y} < y^r$ (which implies that $\theta > \delta$)

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- Implicit function theorem implies $\frac{d\tau^p}{d\theta} = \frac{1}{C''(\tau^p)(1-\delta)} > 0$
- Implicit function theorem implies $\frac{d\tau^p}{d\delta} = \frac{C'(\tau^p)-1}{C''(\tau^p)(1-\delta)} < 0$

Next lecture:

- Nondemocratic politics
- Theory of democratization